

Chapter 14: Flatworms, Mesozoans, and Ribbon Worms

Introduction to Bilateral Animals

- **Bilateral symmetry** (两侧对称): The body can be divided along only one plane of symmetry to yield two mirror-image halves.
- **Cephalization** (头部形成): Concentration of sense organs and nervous control centers on the head (anterior) end.
- **Triploblastic** (三胚层): Body derived from three germ layers: **ectoderm** (外胚层), **endoderm** (内胚层), and **mesoderm** (中胚层).
- **Acoelomate** (无体腔): Animals that lack a coelom (body cavity); the region between the epidermis and digestive cavity is filled with **parenchyma** (实质) (mesodermally derived packing tissue).
- **Protostomia** (原口动物): Clade of animals with spiral or centrolecithal cleavage, mosaic development, and the blastopore becoming the mouth.
- **Lophotrochozoa** (冠轮动物): A clade within Protostomia including Platyhelminthes, Nemertea, Mollusca, Annelida, etc., often sharing a **trochophore** (担轮幼虫) larva or **lophophore** (触手冠).

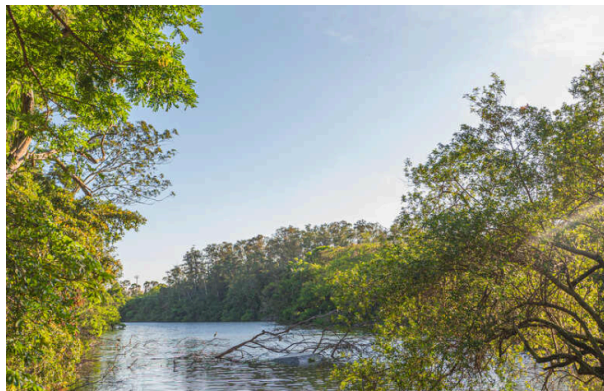


Figure 1: Diagram of acoelomate body plan (cross section).

- **Acoelomate body plan:** As shown in the figure, the gut (endoderm) is surrounded by a solid mass of parenchyma (mesoderm), with no fluid-filled body cavity between the gut and the body wall (ectoderm).

Phylum Acoelomorpha (无腔动物门)

- Small flatworms, less than 5 mm long, typically living in marine sediments.
- Formerly placed in class Turbellaria but now considered an early diverging bilateral group.
- **Statocysts** (平衡囊) differ in structure from those of platyhelminths.
- **Digestion:** Mouth leads to a tube-like pharynx and a sacklike gut (no anus). In many acoels, the gut is absent, and food is digested in a syncytial mass of endodermal cells.
- **Reproduction:** Monoecious; produce **endolecithal eggs** (内卵黄卵); exhibit a unique **duet-spiral cleavage** (二重螺旋卵裂) pattern.



Figure 2: Acoelomorph worms, **Waminoa** sp., on a bubble coral, **Plerogyra sinuosa**.

- **Acoelomorphs**: These small worms, like **Waminoa**, are often found on corals or in sediments.

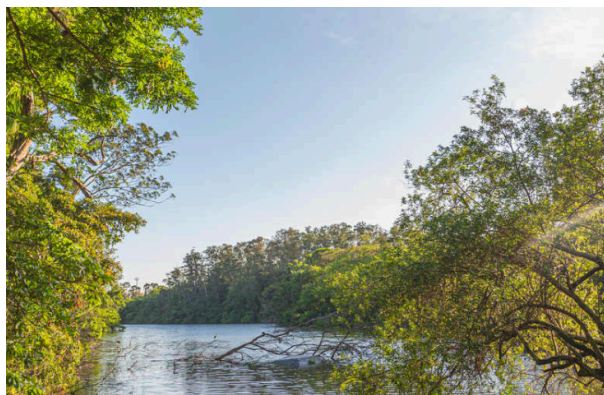


Figure 3: A, Generalized acoelomorph flatworm. B, Midsagittal section showing gut cavity filled with endodermal cells.

- **Acoelomorph structure**: The diagram illustrates the simple body plan with a mouth opening into a mass of endodermal cells rather than a distinct gut cavity in some forms.

Phylum Platyhelminthes (扁形动物门)

- Bilaterally symmetrical, triploblastic, acoelomate animals.
- Flattened dorsoventrally.
- Divided into four classes: **Turbellaria** (涡虫纲) (mostly free-living), **Trematoda** (吸虫纲) (flukes), **Monogenea** (单殖吸虫纲) (monogenetic flukes), and **Cestoda** (绦虫纲) (tapeworms).
- **Neodermata** (新皮类): A clade containing the parasitic classes (Trematoda, Monogenea, Cestoda), characterized by a **syncytial tegument** (合胞体体壁/皮层) (neodermis).



Figure 4: A, Stained planarian. B, **Bipalium**, a terrestrial flatworm.

- **Free-living flatworms:** Planarians (aquatic) and **Bipalium** (terrestrial) are examples of the class Turbellaria.

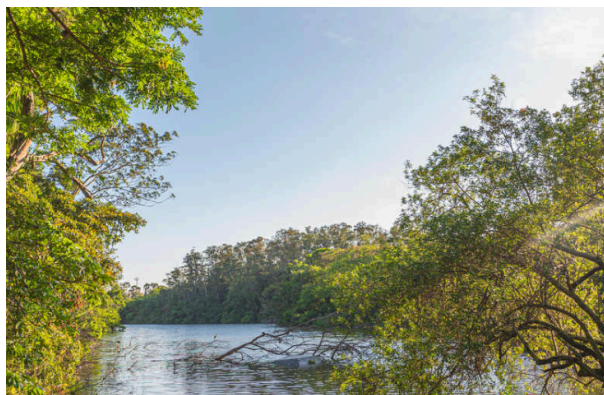


Figure 5: Hypothetical relationships among parasitic Platyhelminthes. The traditionally accepted class Turbellaria is paraphyletic. Some turbellarians have ectolecithal development and, together with the Trematoda, Monogenea, and Cestoda, form a clade and a sister group of the endolecithal turbellarians.

- **Phylogeny:** The cladogram shows that Turbellaria is paraphyletic. The parasitic groups (Trematoda, Monogenea, Cestoda) form a monophyletic clade called **Neodermata**, sharing the syncytial tegument.

Form and Function

Epidermis and Muscles

- **Turbellaria:** Cellular, ciliated epidermis with **rhabdites** (杆状体) (rod-shaped structures that form a protective mucous sheath).
- **Dual-gland adhesive organs** (双腺粘附器官): Consist of viscid gland cells (fasten to substrate), releasing gland cells (detach), and anchor cells.



Figure 6: Cross section of planarian through pharyngeal region, showing relationships of body structures.

- **Planarian cross section:** Shows the ciliated epidermis, rhabdites, and the arrangement of circular, longitudinal, and parenchymal muscles.

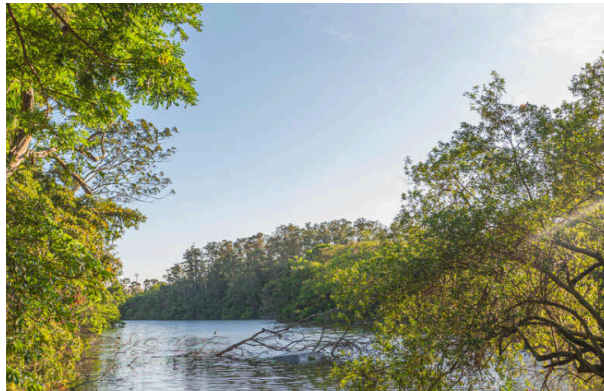


Figure 7: Reconstruction of dual-gland adhesive organ of the turbellarian **Haplopharynx** sp. There are two viscid glands and one releasing gland, which lie beneath the body wall. The anchor cell lies within the epidermis, and one of the viscid glands and the releasing gland are in contact with a nerve.

- **Dual-gland organ:** Illustrates the mechanism of attachment and detachment in turbellarians using viscid and releasing glands.
- **Parasitic classes (Neodermata):** Adults have a non-ciliated **syncytial tegument** (合胞体壁) (neodermis).
- **Syncytial** (合胞体): Many nuclei enclosed within a single cell membrane.
- Tegument is resistant to host immune responses and digestive enzymes.



Figure 8: Diagrammatic drawing of the structure of the tegument of a trematode **Fasciola hepatica**.

- **Tegument structure:** The syncytial tegument has cell bodies sunk beneath the muscle layer, connected to the surface cytoplasm by extensions.

Nutrition and Digestion

- Digestive system typically includes a mouth, pharynx, and intestine (incomplete gut, no anus).
- **Planarians:** Pharynx enclosed in a pharyngeal sheath; intestine has three branches (one anterior, two posterior).
- Digestion is both extracellular (in intestine) and intracellular (gastrodermal phagocytosis).
- **Cestodes:** Lack a digestive system; absorb nutrients through the tegument.

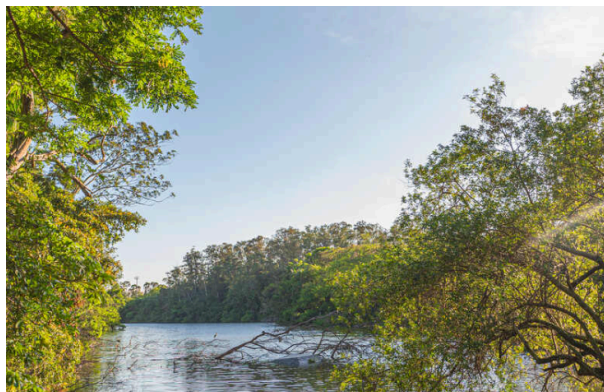


Figure 9: Structure of a planarian. A, Reproductive and osmoregulatory systems, shown in part. Inset at left is enlargement of flame cell. B, Digestive tract and ladder-type nervous system. Pharynx is shown in resting position. C, Pharynx extended through ventral mouth.

- **Planarian systems:**
 - **Digestive:** Three-branched intestine and extensible pharynx.
 - **Excretory:** Protonephridia with flame cells.
 - **Nervous:** Ladder-type system with anterior cerebral ganglia.
 - **Reproductive:** Monoecious with complex gonads and ducts.

Excretion and Osmoregulation

- **Protonephridia** (原肾管): Excretory/osmoregulatory organs with **flame cells** (焰细胞).
- **Flame cell:** Cup-shaped with a tuft of flagella driving fluid through a **weir** (堰) system into collecting ducts.

- Primarily osmoregulatory in freshwater forms; metabolic wastes largely removed by diffusion.

Nervous System and Sense Organs

- **Subepidermal nerve plexus** (表皮下神经丛) (primitive) or **longitudinal nerve cords** (纵神经索) (1-5 pairs).
- **Ladder-type nervous system** (梯状神经系统): Ventral nerve cords connected by transverse nerves.
- **Brain**: Bilobed mass of ganglion cells.
- **Sense organs**: **Ocelli** (眼点) (eyespots), tactile cells, chemoreceptors, **statocysts** (平衡囊) (equilibrium), **rheoreceptors** (流速感受器) (water current).

Reproduction

- Asexual reproduction: Fission (constriction behind pharynx) in planarians; budding in some tapeworms.
- Sexual reproduction: Most are **monoecious** (雌雄同体) (hermaphroditic) with cross-fertilization.
- **Endolecithal eggs** (内卵黄卵): Yolk contained within the egg cell (ancestral).
- **Ectolecithal eggs** (外卵黄卵): Yolk contributed by separate **vitellaria** (卵黄腺) (yolk glands); characteristic of Trematoda, Monogenea, Cestoda, and some Turbellaria.

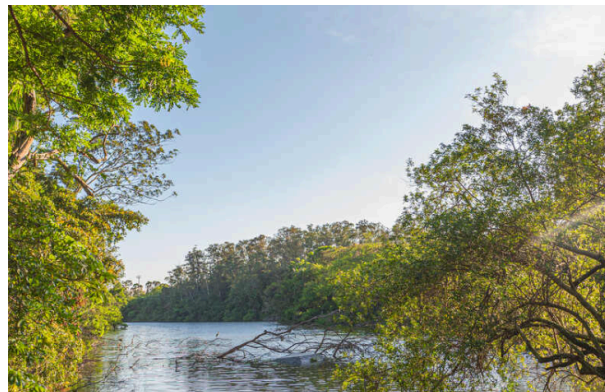


Figure 10: Structure of human liver fluke **Clonorchis sinensis**.

- **Clonorchis reproductive system**: Shows the complex hermaphroditic system typical of flukes, including testes, ovary, uterus, seminal receptacle, and vitellaria.

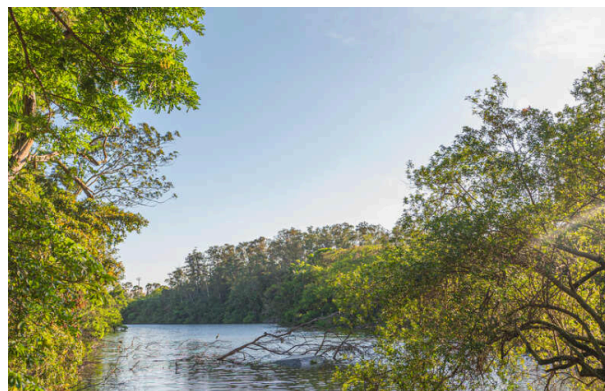


Figure 11: Some small freshwater turbellarians. A, **Phagocata** has numerous pharynges. B and C, Incomplete fission results for a time in a series of attached zooids.

- **Asexual reproduction:** Freshwater turbellarians like **Microstomum** and **Stenostomum** can reproduce by fission, forming chains of zooids.

Class Turbellaria (涡虫纲)

- Mostly free-living; marine, freshwater, or terrestrial.
- Distinguished by gut form and pharynx type.
- **Polycladida** (多肠目): Folded pharynx, highly branched intestine (marine).
- **Tricladida** (三肠目): Three-branched intestine (freshwater planarians).

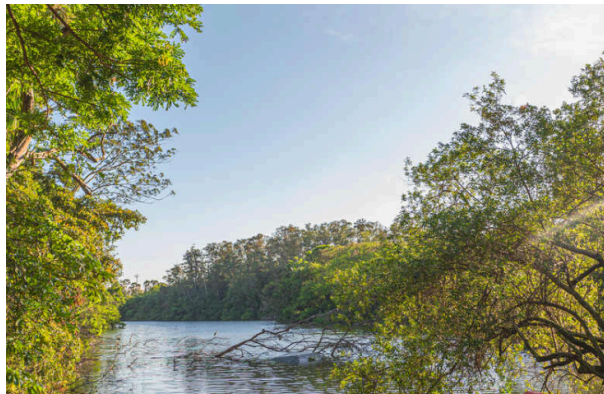


Figure 12: Intestinal pattern of two orders of turbellarians. A, Tricladida. B, Polycladida.

- **Intestinal patterns:** Comparison of the three-branched intestine of Tricladida and the multi-branched intestine of Polycladida.

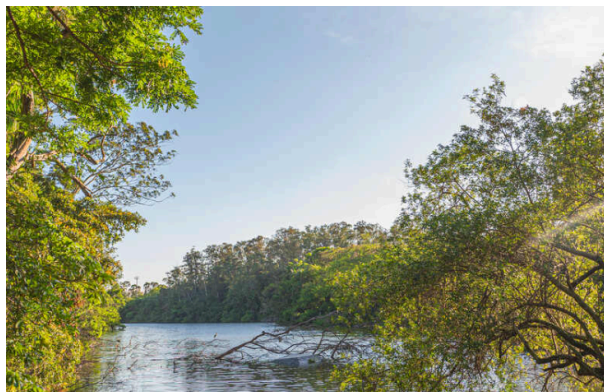


Figure 13: **Pseudobiceros hancockanus**, a marine polyclad turbellarian. Marine polyclads are often large and beautifully colored.

- **Marine polyclad:** **Pseudobiceros** displays the vibrant colors and flat body typical of marine polyclads.

Class Trematoda (吸虫纲)

- All parasitic flukes; adults are endoparasites of vertebrates.
- Leaflike body with suckers (oral and ventral) but no hooks (unlike Monogenea).
- **Subclass Digenea** (复殖亚纲): Complex life cycle with an intermediate host (mollusc) and a definitive host (vertebrate).

Life Cycle Stages

1. **Egg:** Shelled embryo passed in feces.
2. **Miracidium** (毛蚴): Free-swimming ciliated larva; penetrates snail.

3. **Sporocyst** (胞蚴): Sac-like asexual stage in snail.
4. **Redia** (雷蚴): Asexual stage with a gut; produces more rediae or cercariae.
5. **Cercaria** (尾蚴): Free-swimming larva emerging from snail.
6. **Metacercaria** (囊蚴): Encysted juvenile fluke (infective stage).
7. **Adult**: Sexual stage in definitive host.



Figure 14: Life cycle of **Clonorchis sinensis**.

- **Clonorchis life cycle**: Egg → Snail (Miracidium → Sporocyst → Redia → Cercaria) → Fish (Metacercaria) → Human (Adult in liver bile duct).



Figure 15: Examples of Flukes Infecting Humans.

- **Human flukes**: Table lists important species like **Schistosoma**, **Clonorchis**, **Paragonimus**, and **Fasciolopsis**, their distribution, and means of infection.

Important Trematodes

- **Clonorchis sinensis** (华支睾吸虫/肝吸虫) (Chinese liver fluke): Infection via raw fish.
- **Schistosoma** (血吸虫) (Blood flukes): Dioecious (separate sexes); male has **gynecophoric canal** (抱雌沟). Infection via cercariae penetrating skin. No metacercaria stage.
 - **S. mansoni**: Large intestine veins.
 - **S. japonicum**: Small intestine veins.

- **S. haematobium**: Urinary bladder veins.

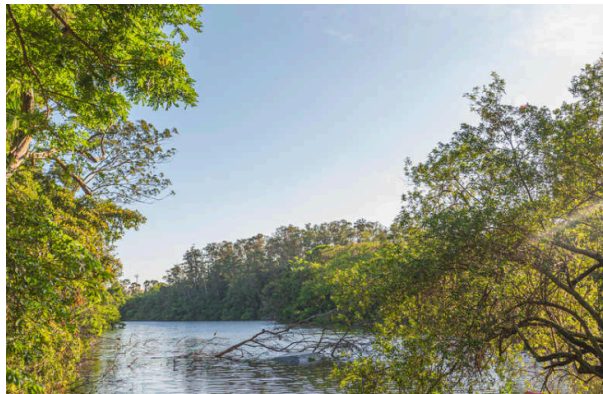


Figure 16: A, Adult male and female **Schistosoma japonicum** in copulation. The male has a long gynecophoric canal that holds the female. B, Life cycle of **Schistosoma mansoni**.

- **Schistosoma**:

- A: Male holding female in gynecophoric canal.
- B: Life cycle involving snail host and direct skin penetration by cercariae.

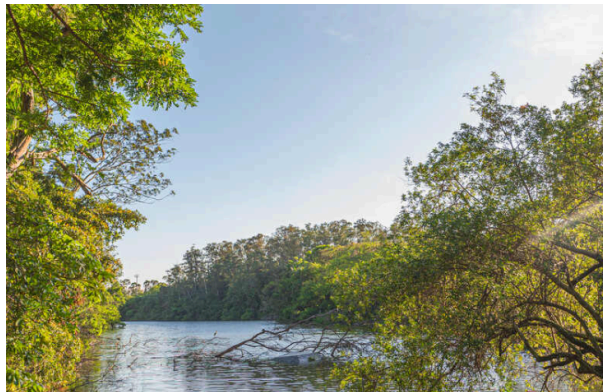


Figure 17: Cut surface of a liver showing severe fibrosis. The patient was a 27-year-old man who died from hematemesis (vomiting blood) associated with spleen and liver enlargement. Over 180 pairs of adult **Schistosoma mansoni** were counted at autopsy.

- **Pathology**: Schistosomiasis can cause severe liver fibrosis due to immune reaction against eggs trapped in tissues.
- **Paragonimus westermani** (卫氏并殖吸虫/肺吸虫) (Lung fluke): Infection via raw crabs/ crayfish.
- **Leucochloridium** (彩蚴吸虫): Sporocysts enter snail tentacles and pulsate to attract bird hosts.



Figure 18: Lung fluke **Paragonimus westermani**. Adults are up to 2 cm long. Eggs discharged in sputum or feces hatch into free-swimming miracidia that enter snails. Cercariae from snails enter freshwater crabs and encyst in soft tissues. Humans are infected by eating poorly cooked crabs or by drinking water containing larvae freed from dead crabs.

- **Paragonimus**: Adult fluke lives in lungs; eggs coughed up or swallowed.

Class Monogenea (单殖吸虫纲)

- Ectoparasites, primarily on fish gills and skin.
- Direct life cycle (single host).
- **Opisthaptor** (后吸器): Large posterior attachment organ with hooks, suckers, or clamps.
- **Oncomiracidium** (钩毛蚴): Ciliated larva.

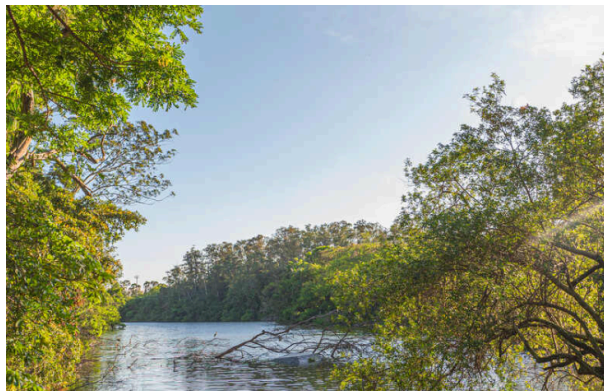


Figure 19: A monogenetic fluke **Gyrodactylus cylindriciformis**, ventral view.

- **Monogenean structure**: **Gyrodactylus** showing the opisthaptor with hooks for attachment to the host.

Class Cestoda (绦虫纲)

- Endoparasites in vertebrate digestive tracts.
- Long, flat body composed of:
 - **Scolex** (头节): Holdfast organ with suckers/hooks.
 - **Neck** (颈部): Germinative zone.
 - **Strobila** (链体): Chain of **proglottids** (节片) (reproductive units).
- No digestive system.
- **Microtriches** (微毛): Minute projections on tegument to increase surface area for absorption.



Figure 20: A tapeworm, showing strobila and scolex. The scolex is the organ of attachment.

- **Tapeworm body plan:** Scolex for attachment, followed by a strobila of immature, mature, and gravid proglottids.

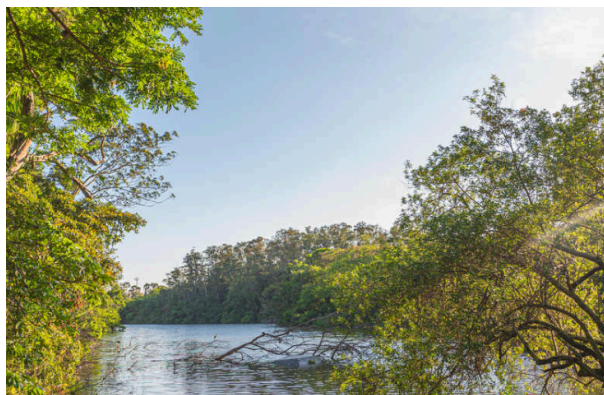


Figure 21: Schematic drawing of a longitudinal section through a sensory ending in the tegument of **Echinococcus granulosus**.

- **Tegument sensory ending:** Shows the microtriches (microvilli-like) and a sensory cilium.

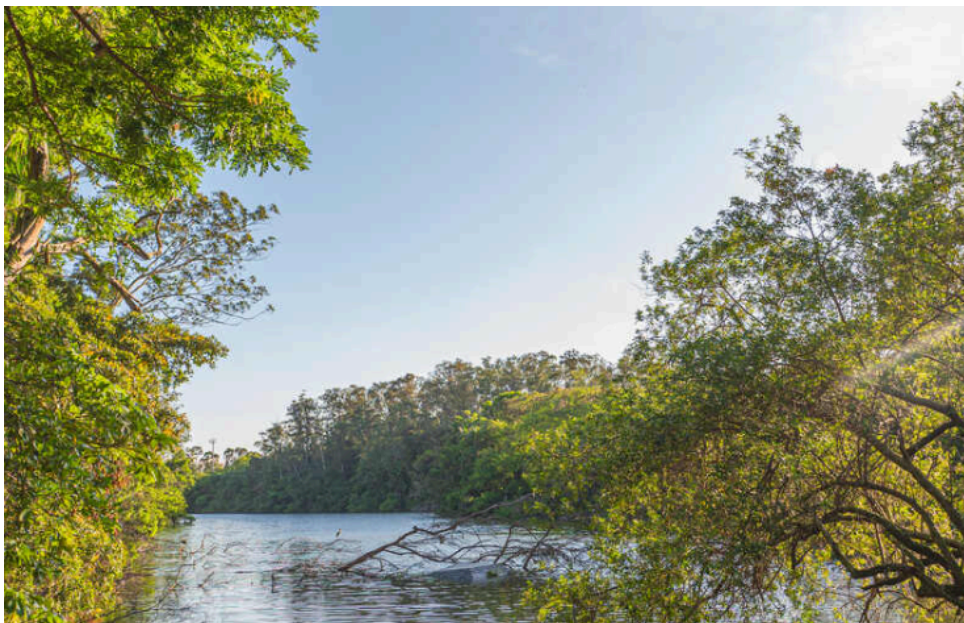


Figure 22: Common Cestodes of Humans.

- **Human Cestodes:** Lists *Taenia saginata*, *Taenia solium*, *Diphyllobothrium latum*, *Echinococcus granulosus*, etc.

Important Cestodes

- ***Taenia saginata*** (牛带绦虫) (Beef tapeworm): Scolex with 4 suckers, no hooks. Cysticerci in cattle muscle (“measly beef”).
- ***Taenia solium*** (猪带绦虫) (Pork tapeworm): Scolex with suckers and hooks (**rostellum** (顶突)). Cysticerci in pig muscle. Dangerous because humans can serve as intermediate hosts (**cysticercosis** (囊尾蚴病)).



Figure 23: Life cycle of beef tapeworm, ***Taenia saginata***. Ripe proglottids detach in the human intestine, leave the body in feces, crawl onto grass, and are ingested by cattle. Eggs hatch in the cow’s intestine, freeing oncospheres, which penetrate into muscles and encyst, developing into “bladder worms.” A human eats infected rare beef, and cysticercus is freed in intestine where it attaches to the intestinal wall, forms a strobila, and matures.

- ***Taenia saginata* life cycle:** Human (definitive host) → Eggs in feces → Cattle (intermediate host) → Cysticercus in muscle → Human eats rare beef.

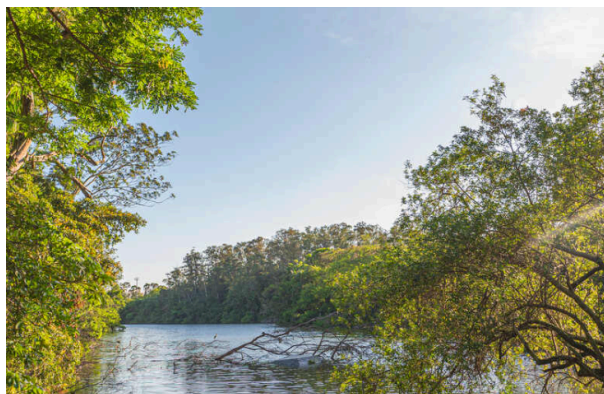


Figure 24: Mature proglottid of ***Taenia pisiformis***, a dog tapeworm. Portions of two other proglottids also shown.

- **Mature proglottid:** Contains complete male (testes, vas deferens) and female (ovary, uterus, vitelline gland) reproductive systems.



Figure 25: Section through the brain of a person who died of cerebral cysticercosis, an infection with cysticerci of **Taenia solium**.

- **Cysticercosis**: Brain section showing infection with **Taenia solium** cysticerci, a potentially fatal condition.
- **Diphyllobothrium latum** (阔节裂头绦虫) (Fish tapeworm): Largest cestode infecting humans; can cause anemia.
- **Echinococcus granulosus** (细粒棘球绦虫) (Dog tapeworm): Causes **hydatidosis** (棘球蚴病) (hydatid cysts) in humans (intermediate host).

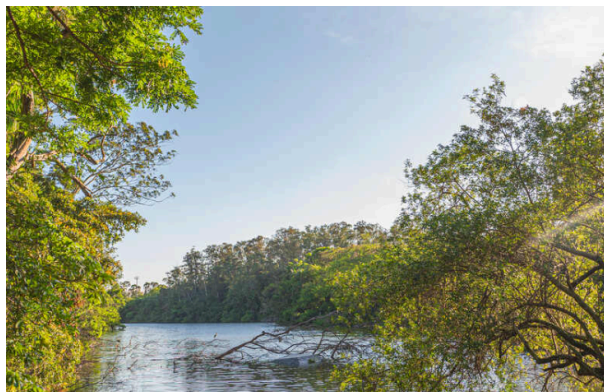


Figure 26: **Echinococcus granulosus**, a dog tapeworm, which may be dangerous to humans. A, Early hydatid cyst or bladder-worm stage found in cattle, sheep, hogs, and sometimes humans produces hydatid disease. B, The adult tapeworm lives in intestine of a dog or other carnivore.

- **Echinococcus**:
 - A: Hydatid cyst with brood capsules containing scolices.
 - B: Small adult worm living in dog intestine.

Phylum Mesozoa (中生动物门)

- Minute, ciliated, wormlike parasites of marine invertebrates.
- Simple organization: 20-30 cells in two layers (not homologous to germ layers).
- **Rhombzoa** (菱形虫纲): Parasites in cephalopod kidneys. Adults called **vermiforms** (蠕虫形).
- **Orthonectida** (直泳虫纲): Parasites of brittle stars, molluscs, etc.



Figure 27: Two methods of reproduction by mesozoans. A, Asexual development of vermiform larvae from reproductive cells in the axial cell of an adult. B, Under crowded conditions in the host kidney, reproductive cells develop into gonads with gametes that produce infusoriform dispersal larvae that emerge in the host urine.

- **Mesozoan reproduction:** Shows asexual production of vermiform larvae and sexual production of infusoriform larvae in Rhombozoa.

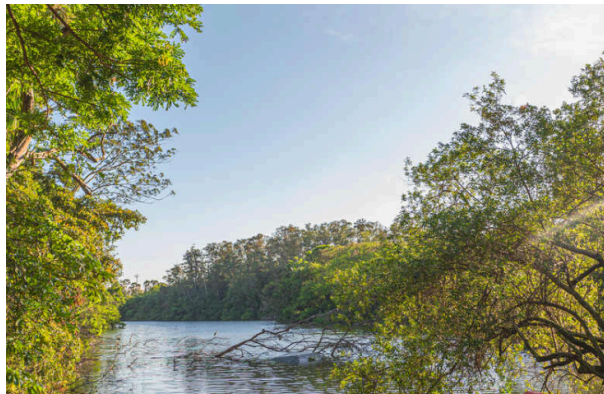


Figure 28: A, Female and, B, male orthonectid (**Rhopalura**). This mesozoan parasitizes such forms as flatworms, molluscs, annelids, and brittle stars. The structure is a single layer of ciliated epithelial cells surrounding an inner mass of sex cells.

- **Orthonectid: Rhopalura** showing the simple structure of ciliated cells surrounding sex cells.

Phylum Nemertea (纽形动物门)

- Also called **Rhynchocoela** (吻腔动物).
- Bilaterally symmetrical, triploblastic.
- **Proboscis** (吻): Long muscular tube in a fluid-filled cavity (**rhynchocoel** (吻腔)) above the gut; eversible to capture prey.
- **Complete digestive system:** Mouth and anus.
- **Circulatory system:** Closed loop system.
- **Reproduction:** Mostly dioecious; some species produce a ciliated **pilidium** (帽状幼虫) larva.



Figure 29: Ribbon worm **Amphiporus bimaculatus** (phylum Nemertea) is 6 to 10 cm long, but other species range up to several meters. The proboscis of this specimen is partially extended at the top; the head is marked by two brown spots. The animal is photographed on an algal frond.

- **Ribbon worm: Amphiporus** showing the extended proboscis.

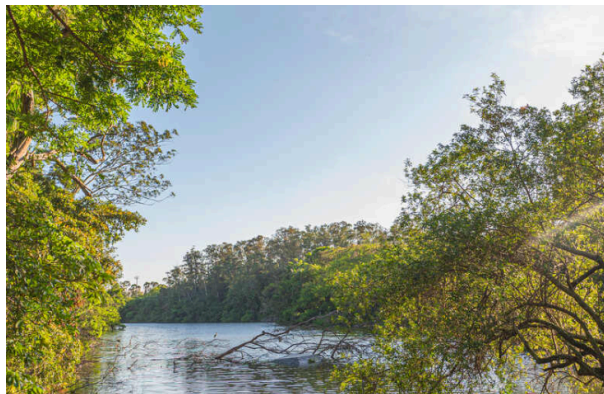


Figure 30: A, **Amphiporus**, with proboscis extended to catch prey. B, Structure of female nemertean worm **Amphiporus** (diagrammatic). Dorsal view to show proboscis.

- **Proboscis mechanism:**
 - A: Proboscis extended to catch prey.
 - B: Diagram showing the rhynchocoel above the digestive tract and the stylet on the proboscis.

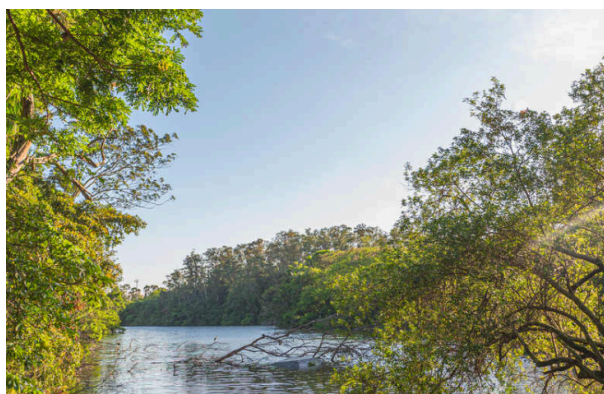


Figure 31: **Baseodiscus** is a genus of nemerteans whose members typically measure several meters in length. This **B. mexicanus** is from the Galápagos Islands.

- **Large nemerteans: Baseodiscus** can reach several meters in length.



Figure 32: A, Diagrammatic cross section of female nemertean worm. B, Excretory and circulatory systems of nemertean worm. Flame bulbs along nephridial canal are closely associated with lateral blood vessels.

- **Nemertean anatomy:**

- A: Cross section showing rhynchocoel, proboscis, and gut.
- B: Excretory (flame bulbs) and circulatory systems closely associated.